Interactions in QFT & Feynman Rules

Now that we've quantized the free scalar field, we'll consider quantum field theories with interactions.

Our goal will be to compute scattering amplitudes, in analogy with our study of ~~time~~ perturbation theory in quantum mechanics.

Relatively soon, we'll see that these amplitudes can be expressed via Feynman diagrams which will give us a fast algorithm to computing them.

Just as in time-dependent perturbation theory in QM where the perturbing Hamiltonian allowed transitions between eigenstates of the unperturbed Hamiltonian, the interactions in QFT will allow for transitions between multiparticle states of the free theory — decays & scattering.

We'll introduce machinery to compute these amplitudes & convert them into measurable quantities like cross sections & decay rates.

ϕ^4 theory

Building off our free particle example, consider the real scalar field theory before, with a new quartic term in its potential energy:

$$\propto \lambda \phi^4$$

The Lagrangian is:

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 - \frac{1}{4!} \lambda \phi^4$$

The factor $4!$ is introduced for later convenience. The prefactor λ is a dimensionless coupling constant.

The Hamiltonian is

$$H = \int d^3x \, \mathcal{H} = \int d^3x \left[\frac{1}{2} \pi^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4 \right]$$

As in QM, our first step is to identify the unperturbed Hamiltonian, H_0:

$$H_0 = \int d^3x \left[\frac{1}{2} \pi^2 + \frac{1}{2} (\vec{\nabla} \phi)^2 + \frac{1}{2} m^2 \phi^2 \right]$$

and the interacting part:

$$H_{int} = \int d^3x \, \frac{\lambda}{4!} \phi^4$$

A First Look at Scattering

Just as in QM, we can work in the small λ limit, and look for transitions between the free particle states that are eigenstates of H_0.

To set up the idea a bit more, consider the scattering process:

$$\phi(p_1) + \phi(p_2) \longrightarrow \phi(p_3) + \phi(p_4)$$

Just as in QM, we want to compute a transition amplitude:

$$\langle f | \left(-i \int dt\, H_{int} \right) | i \rangle$$

where, $|i\rangle$, $|f\rangle$ are the initial / final states. For our example process, they are of the form:

$$|i\rangle = |\vec{p}_1, \vec{p}_2\rangle = \sqrt{2E_{\vec{p}_1}}\sqrt{2E_{\vec{p}_2}}\, a^\dagger_{\vec{p}_1} a^\dagger_{\vec{p}_2} |0\rangle$$

$$\langle f| = \langle \vec{p}_3, \vec{p}_4| = \sqrt{2E_{\vec{p}_3}}\sqrt{2E_{\vec{p}_4}}\, \langle 0 | a_{\vec{p}_3} a_{\vec{p}_4}$$

With H_{int} defined above, the amplitude of interest is thus:

$$\langle \vec{p}_3\, \vec{p}_4 | \left\{ -i \int \underbrace{dt \int d^3x}_{= \int d^4x} \frac{\lambda}{4!} \phi^4(t,x) \right\} | \vec{p}_1, \vec{p}_2 \rangle$$

Since we're working in perturbation theory, we ~~differentiate~~ construct $\phi(x)$ explicitly in terms of the ladder operators:

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \left(a_{\vec{p}} e^{-ipx} + a^\dagger_{\vec{p}} e^{ipx} \right)$$

To start evaluating this, we consider the action of $\phi(x)$ on $|\vec{p}\rangle$:

$$\phi(x)|\vec{p}\rangle = \left(\int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2E_k}} a_k e^{-ik\cdot x} + \dots \right) \left(\sqrt{2E_{\vec{p}}}\, a^\dagger_{\vec{p}} |0\rangle \right)$$

$$= \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{2E_{\vec{p}}}{2E_k}}\, e^{-ik\cdot x} \left(\underbrace{[a_k, a^\dagger_{\vec{p}}]}_{(2\pi)^3 \delta^3(\vec{k}-\vec{p})} - a^\dagger_{\vec{p}} a_k \right) |0\rangle$$

$$= \int d^3k\, e^{-ik\cdot x}\, \delta^3(\vec{k}-\vec{p})\, |0\rangle = e^{-ip\cdot x} |0\rangle$$

Similarly, the action of ϕ on $\langle\vec{p}|$ is:

$$\langle\vec{p}|\phi(x) = \langle 0| e^{+ip\cdot x}$$

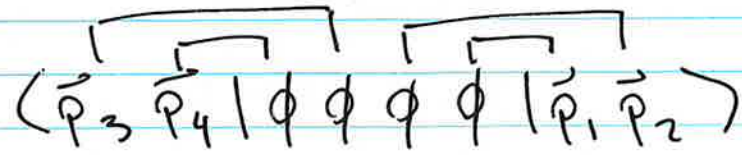
These types of operations are called <u>contractions</u> of the field operator w/ the ~~final~~ external state.

We'll use the notation

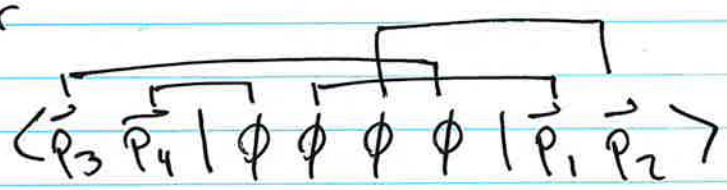
$$\overbrace{\phi(x)|\vec{p}\rangle} \quad \text{&} \quad \overbrace{\langle\vec{p}|\phi(x)}$$

This is useful because the external states are unordered, and we can see all the pairings quickly.

We can contract the four field operators w/ the four final states. For example:



or



Some quick counting reveals there are 4! possible contractions, all leading to the same result.

The transition amplitude is thus:

$$\langle \vec{p}_3 \vec{p}_4 | \left(-i \int d^4x \, \frac{\lambda}{4!} \phi^4(x) \right) | \vec{p}_1 \vec{p}_2 \rangle$$

$$= \frac{-i\lambda}{4!} \times 4! \times \int d^4x \, \langle \vec{p}_3 \vec{p}_4 | \phi \phi \phi \phi | \vec{p}_1 \vec{p}_2 \rangle$$

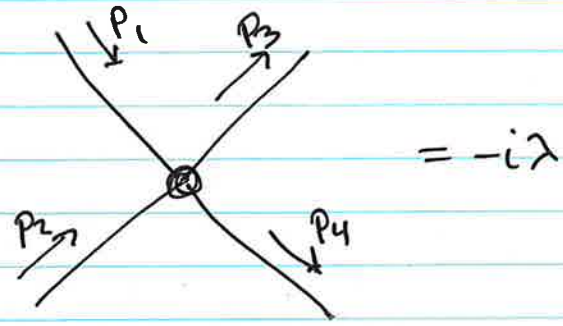
$$= -i\lambda \cdot \int d^4x \, e^{-i(p_1+p_2-p_3-p_4)\cdot x}$$

$$= -i\lambda \, (2\pi)^4 \, \delta^4(p_1+p_2-p_3-p_4)$$

→ this is the transition amplitude!

Note that the transition amplitude includes a δ-fcn. that enforces energy & momentum conservation.

The amplitude can be represented graphically in terms of a Feynman diagram:



the association w/ a factor $-i\lambda$ is the Feynman rule (in momentum space) we will learn to associate w/ this diagram shortly.

Later, we'll return to how this amplitude can be used to compute a cross-section for $\phi\phi \to \phi\phi$ scattering.

First, we'll address several issues we swept under the rug:

→ We used H_0 for the initial & final states, and the vacuum and the mode expansion of the free field in evaluating the matrix element.

→ How do we rigorously justify this in QFT?

Perturbative Calculation of Correlation Functions

Before computing matrix elements between multi-particle states, we'll consider simpler objects — correlation functions, or vacuum-to-vacuum transitions.

Consider the two-point correlation function:

$$\langle \Omega | T\{\phi(x)\phi(y)\} | \Omega \rangle$$

Here, $|\Omega\rangle$ is the vacuum state in the interacting theory, (to distinguish it from the vacuum of the free theory, $|0\rangle$)

T represents time-ordering, as before.

This correlation function schematically represents the amplitude of a particle to propagate from y to x.

In the free theory, it's given by the Feynman propagator:

$$\langle 0 | T\{\phi(x)\phi(y)\} | 0 \rangle$$

$$= D_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip\cdot(x-y)}$$

We want to see how this changes in the interacting theory. The Hamiltonian of ϕ^4 theory is:

$$H = H_0 + H_{int} = H_0 + \int d^3x \frac{\lambda}{4!}\phi^4(\vec{x})$$

Here, $\phi(\vec{x})$ is a Schrödinger picture field, while $\phi(x)$ in the correlator is a Heisenberg field.

We want to compute the full correlator as a power series in λ. The interacting part of the theory appears in two places: first, in the definition of the Heisenberg field:

$$\phi(x) = e^{iHt}\,\phi(\vec{x})\,e^{-iHt},$$

and second, in the definition of $|\Omega\rangle$.

We'll relate both these quantities to free field operators and the free theory vacuum, $|0\rangle$, which we know how to work with.

Let's start with $\phi(x)$. At any fixed time t_0, we can express ϕ as:

$$\phi(t_0,\vec{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{p}}}}\left(a_{\vec{p}}\,e^{i\vec{p}\cdot\vec{x}} + a^{\dagger}_{\vec{p}}\,e^{-i\vec{p}\cdot\vec{x}}\right)$$

Then, to obtain $\phi(t,\vec{x})$ for $t \neq t_0$, we just move to the Heisenberg picture:

$$\phi(t,\vec{x}) = e^{iH(t-t_0)}\,\phi(t_0,\vec{x})\,e^{-iH(t-t_0)}$$

For $\lambda = 0$, $H = H_0$ and this reduces to:

$$\phi(t,\vec{x})\Big|_{\lambda=0} = e^{iH_0(t-t_0)}\,\phi(t_0,\vec{x})\,e^{-iH_0(t-t_0)}$$

$$\equiv \phi_I(t,\vec{x})$$

where we have defined the <u>interaction picture</u> field,

$$\phi_I(t,\vec{x}) = \phi_I(x).$$

This operator is particularly convenient because we know how to construct it explicitly:

$$\phi_I(t,\vec{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{p}}}}\left(a_{\vec{p}}\,e^{-ipx} + a^\dagger_{\vec{p}}\,e^{ip\cdot x}\right)_{x^0=t-t^0}$$

Now we can express the <u>full</u> Heisenberg field in terms of ϕ_I:

$$\phi(t,\vec{x}) = e^{iH(t-t_0)}\underbrace{\left[e^{-iH_0(t-t_0)}\,\phi_I(t,\vec{x})\,e^{+iH_0(t-t_0)}\right]}_{\text{the Schrödinger field}}e^{-iH(t-t_0)}$$

$$\equiv U^\dagger(t,t^0_0)\,\phi_I(t,\vec{x})\,U(t,t_0)$$

where we've defined the unitary operator

$$U(t,t_0) = e^{iH_0(t-t_0)}\,e^{-iH(t-t_0)}$$

This is the <u>time-evolution operator</u> in the interaction picture.

Now we want to write $U(t,t_0)$ in terms of ϕ_I, since we have the latter already expressed in terms of ladder operators.

To this end, note that $U(t,t_0)$ solves the Schrödinger equation, and is in fact the unique solution w/ initial condition $U(t_0,t_0)=1$.

$$i\frac{\partial}{\partial t}U(t,t_0) = i\frac{\partial}{\partial t}\left[e^{iH_0(t-t_0)}e^{-iH(t-t_0)}\right]$$

for an operator, $\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A$ (A commutes w/ itself)

$$\Rightarrow \quad = e^{iH_0(t-t_0)}(H-H_0)e^{-iH(t-t_0)}$$

$$= e^{iH_0(t-t_0)}H_{int}e^{-iH(t-t_0)}$$

$$= e^{iH_0(t-t_0)}H_{int}e^{-iH_0(t-t_0)}\underbrace{e^{iH_0(t-t_0)}e^{-iH(t-t_0)}}_{U(t,t_0)}$$

$$\equiv H_I(t)U(t,t_0)$$

where we've defined the perturbation in the interaction picture:

$$H_I(t) \equiv e^{iH_0(t-t_0)}H_{int}e^{-iH_0(t-t_0)}$$

$$= e^{iH_0(t-t_0)}\int d^3x\,\frac{\lambda}{4!}\phi^4(t_0,\vec{x})e^{-iH_0(t-t_0)}$$

$$= \int d^3x\,\frac{\lambda}{4!}\phi_I^4(x)$$

In the last step, we inserted several factors of $e^{-iH_0(t-t_0)}\, e^{iH_0(t-t_0)} = 1$ b/w the Schrödinger picture fields to get to $\phi_I(x)$.

Now we can solve the differential equation as a series in the small parameter λ. The solution is:

$$U(t,t_0) = 1 + (-i)\int_{t_0}^{t} dt_1\, H_I(t_1) + (-i)^2 \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2\, H_I(t_1) H_I(t_2)$$
$$+ (-i)^3 \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \int_{t_0}^{t_2} dt_3\, H_I(t_1) H_I(t_2) H_I(t_3) + \dots$$

To check this is the solution, just compute $i\frac{\partial}{\partial t} U(t,t_0)$. Each term gives the previous one, times $H_I(t)$. Factor out $H_I(t)$ in front to get $H_I U(t,t_0)$. The initial condition $U(t_0,t_0) = 1$ is clearly satisfied (all integrals vanish)

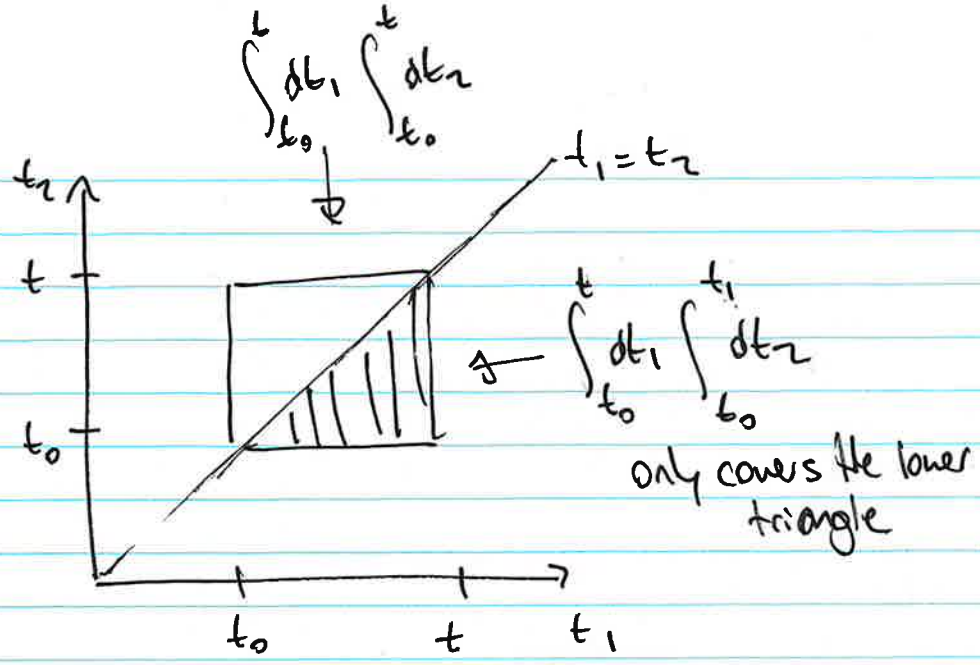
Note that each term in $U(t,t_0)$ has the factors of $H_I(t)$ arranged in <u>time order</u> — later on the left ($t_1 > t_2 > t_3 > \dots$)

Thus, for instance, we can write

$$\int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2\, H_I(t_1) H_I(t_2) = \frac{1}{2}\int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2\, T\left[H_I(t_1) H_I(t_2)\right]$$

(now t_1, not t_1!)

this works b/c the integrand on the RHS is symmetric about $t_2 = t_1$ — the integral on the LHS only covers the lower triangle in the following figure:



This generalizes to higher orders:

$$\int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n \, H_I(t_1) \cdots H_I(t_n)$$

$$= \frac{1}{n!} \int_{t_0}^{t} dt_1 \cdots \int_{t_0}^{t} dt_n \, T\{H_I(t_1) \cdots H_I(t_n)\}$$

Thus we can write:

$$U(t, t_0) = 1 + (-i) \int_{t_0}^{t} dt_1 \, H_I(t_1) + \frac{(-i)^2}{2} \int_{t_0}^{t} dt_1 \int_{t_0}^{t} dt_2 \, T[H_I(t_1) H_I(t_2)] + \cdots$$

$$= T\left\{ \exp\left[-i \int_{t_0}^{t} dt' \, H_I(t') \right] \right\}$$

This is known as the Dyson series.

To recap, we have shown how to express the Heisenberg fields (which appear in the correlator of interest) in terms of the interaction field, $\phi_I(x)$:

$$\phi_I(x) = U^\dagger(t, t_0)\, \phi_I(x)\, U(t, t_0)$$

w/ $U(t, t_0)$ expressed in terms of ϕ_I via $H_I(t)$.

Before proceeding, note several useful properties of $U(t, t')$ (where t' is now arbitrary, rather than the time we fixed our Schrödinger fields):

$$U(t, t') = T\exp\left[-i\int_{t'}^{t} dt''\, H_I(t'')\right] \quad (t > t')$$

$$i\frac{\partial}{\partial t} U(t, t') = H_I(t)\, U(t, t') \;;\; U(t, t) = 1$$

and

$$U(t, t') = e^{iH_0(t-t_0)}\, e^{-iH(t-t')}\, e^{-iH_0(t'-t_0)}$$

Note that this implies that U is unitary: $U^\dagger = U^{-1}$.

Finally, U satisfies:

$$U(t_1, t_2)\, U(t_2, t_3) = U(t_1, t_3)$$

$$U(t_1, t_3)\left(U(t_2, t_3)\right)^\dagger = U(t_1, t_2)$$

Now we turn to the vacuum state $|\Omega\rangle$.

We want to relate this to the free vacuum state $|0\rangle$ that we know how to compute with. Consider the action of e^{-iHT} on $|0\rangle$, inserting a complete set of energy eigenstates:

$$e^{-iHT}|0\rangle = e^{-iHT}\sum_n |n\rangle\langle n|0\rangle = \sum_n e^{-iE_nT}|n\rangle\langle n|0\rangle$$

w/ E_n the energy eigenvalues of H. (This is an enormously degenerate, continuous sum, but that doesn't matter for our purposes)

We must assume - reasonably - that $\langle\Omega|0\rangle \neq 0$ (otherwise H_{int} could hardly be a "small" perturbation)

Then, the series above contains $|\Omega\rangle$:

$$e^{-iHT}|0\rangle = e^{-iE_0T}|\Omega\rangle\langle\Omega|0\rangle + \sum_{n\neq 0} e^{-iE_nT}|n\rangle\langle n|0\rangle$$

w/ $E_0 = \langle\Omega|H|\Omega\rangle$ is the ground state energy.

We shift the zero of energy s.t. $H_0|0\rangle = 0$. Then, since $E_n > E_0$, we can remove the terms w/ $n \neq 0$ in the series by taking the large T limit with a small imaginary part:

$$T \to \infty(1 - i\epsilon)$$

Then all the exponentials have a small damping component, but this is smallest for $n=0$ $(e^{-\epsilon E_n T})$

So we have

$$|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \left(e^{-iE_0 T} \langle \Omega | 0 \rangle \right)^{-1} e^{-iHT} |0\rangle$$

Now for large T, we can shift $T \to T + t_0$ so that

$$|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \left(e^{-iE_0(T+t_0)} \langle \Omega | 0 \rangle \right)^{-1} e^{-iH(T+t_0)} |0\rangle$$

insert $1 = e^{-iH_0(-T-t_0)}$

then, looking back at our expression for $U(t,t')$ we recognize

$$e^{iH(-T-t_0)} e^{-iH_0(-T-t_0)} = U^\dagger(-T, t_0) = U(t_0, -T)$$

So

$$|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \underbrace{\left(e^{-iE_0(t-(-T))} \langle \Omega | 0 \rangle \right)^{-1}}_{\text{this is some constant prefactor}} U(t_0, -T)|0\rangle$$

otherwise, we see we obtain $|\Omega\rangle$ by evolving $|0\rangle$ from time $-T$ to t_0 w/ $U(t_0, -T)$. A similar calculation gives:

$$\langle \Omega | = \lim_{T\to\infty(1-i\epsilon)} \langle 0 | U(T, t_0) \left(e^{-iE_0(T-t_0)} \langle 0 | \Omega \rangle \right)^{-1}$$

Now we ~~see~~ have all the pieces we need to compute the correlation function.

Let's assume $x^0 > y^0$. Then,

$$\langle \Omega | \phi(x)\phi(y) | \Omega \rangle = \lim_{T \to \infty(1-i\epsilon)} \left(e^{-iE_0(T-t_0)} \langle 0|\Omega\rangle \right)^{-1}$$

$$\times \langle 0 | U(T,t_0) \left\{ \left[U(x^0,t^0) \right]^\dagger \phi_I(x) U(x^0,t_0) \right\}$$

$$\times \left\{ \left[U(y^0,t_0) \right]^\dagger \phi_I(y) U(y^0,t_0) \right\} U(t_0,-T)|0\rangle$$

$$\times \left(e^{-iE_0(t_0-(-T))} \langle 0|\Omega\rangle \right)^{-1}$$

$$= \lim_{T \to \infty(1-i\epsilon)} \left(|\langle 0|\Omega\rangle|^2 e^{-iE_0(2T)} \right)^{-1}$$

$$\times \langle 0 | U(T,x^0) \phi_I(x^0) U(x^0,y^0) \phi_I(y) U(y^0,-T)|0\rangle$$

Now we divide this by 1 in the form

$$1 = \langle \Omega | \Omega \rangle$$

$$= \lim_{T \to \infty(1-i\epsilon)} \left(|\langle 0|\Omega\rangle|^2 e^{-iE_0(2T)} \right)^{-1} \langle 0 | \underbrace{U(T,t_0)U(t_0,-T)}_{= \langle 0|U(T,-T)|0\rangle} |\Omega\rangle$$

Putting it all together:

$$\langle \Omega | \phi(x) \phi(y) | \Omega \rangle =$$

$$\lim_{T \to \infty(1-i\epsilon)} \frac{\langle 0 | U(T, x^0) \phi_I(x) U(x^0, y^0) \phi_I(y) U(y^0, -T) | 0 \rangle}{\langle 0 | U(T, -T) | 0 \rangle}$$

Note that all the operators are time ordered. This would've happened if $y^0 > x^0$ too, so we can rewrite this w/ the time ordering symbol on each side:

$$\langle \Omega | T\{\phi(x)\phi(y)\} | \Omega \rangle$$

$$= \lim_{T \to \infty(1-i\epsilon)} \frac{\langle 0 | T\{\phi_I(x)\phi_I(y) \exp\left[-i\int_{-T}^{T} dt\, H_I(t)\right]\} | 0 \rangle}{\langle 0 | T\{\exp\left[-i\int_{-T}^{T} dt\, H_I(t)\right]\} | 0 \rangle}$$

A similar formula holds for more factors of ϕ put into the correlator.

This is a big step in what we'll eventually need for scattering amplitude calculations.

~~Next, we'll need to learn how~~

For practical calculations, we keep only the first few terms in the expansion of the exponent.

Now we'll see how to handle the resulting terms efficiently.

Wick's Theorem

To make further progress, we need to evaluate expressions of the form:

$$\langle 0 | T\{ \phi_I(x_1)\phi_I(x_2)\dots\phi_I(x_n)\} | 0 \rangle$$

— time-ordered vacuum expectation values of (interaction picture) free field operators.

For $n=2$, this is the Feynman propagator:

$$\langle 0 | T\{ \phi_I(x_1)\phi_I(x_2)\} | 0 \rangle = D_F(x_1 - x_2)$$

* From here on, I'll drop the subscript I, but remember we're always dealing w/ interaction picture field operators. *

Start w/ the case of two fields,

$$\langle 0 | T\{ \phi(x)\phi(y)\} | 0 \rangle$$

To help generalize this, split ϕ into positive & negative frequency parts:

$$\phi(x) = \phi^+(x) + \phi^-(x)$$

w/

$$\phi^+(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} a_{\vec{p}}\, e^{-ip\cdot x}$$

$$\phi^-(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} a^\dagger_{\vec{p}}\, e^{ip\cdot x}$$

This is useful because

$$\phi^+(x)|0\rangle = 0 \quad , \quad \langle 0|\phi^-(x) = 0$$

Now, assume ~~that~~ first that $x^0 > y^0$. Then,

$$T\{\phi(x)\phi(y)\} \underset{x^0>y^0}{=} \phi^+(x)\phi^+(y) + \phi^+(x)\phi^-(y) + \phi^-(x)\phi^+(y) + \phi^-(x)\phi^-(y)$$

$$= \phi^+(x)\phi^+(y) + \phi^-(y)\phi^+(x) + \phi^-(x)\phi^+(y) + \phi^-(x)\phi^-(y) + \left[\phi^+(x)\phi^-(y)\right]$$

In every term except the commutator, the a_p's are positioned to the right of the $a_p^\dagger$'s. Such terms, such as $(a_p^\dagger a_q^\dagger a_r a_s)$ are said to be in <u>normal order</u>, and have vanishing expectation value.

We define the <u>Normal Ordering</u> symbol $N(\ )$ which ~~creates~~ puts the operators in normal order:

$$N(a_p a_r^\dagger a_q) = a_r^\dagger a_p a_q$$

ordering irrelevant since they commute.

(sometimes $N(\ldots)$ is denoted $:\ldots:$)

Next, consider the case $y^0 > x^0$. Then,

$$T\{\phi(x)\phi(y)\} \underset{y^0>x^0}{=} \phi^+(y)\phi^+(x) + \phi^+(y)\phi^-(x) + \phi^-(y)\phi^+(x) + \phi^-(y)\phi^-(x)$$

all normal ordered

$$= \phi^+(y)\phi^+(x) + \phi^-(x)\phi^+(y) + \phi^-(y)\phi^+(x) + \phi^-(y)\phi^-(x) + \left[\phi^+(y), \phi^-(x)\right]$$

Now we define the contraction of two fields as

$$\overbrace{\phi(x)\phi(y)} = \begin{cases} \left[\phi^+(x), \phi^-(y)\right] & \text{for } x^0 > y^0 \\ \left[\phi^+(y), \phi^-(x)\right] & y^0 \not> x^0 \end{cases}$$

This is, in fact, the Feynman propagator!

$$\overbrace{\phi(x)\phi(y)} = D_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\epsilon} e^{-ip\cdot(x-y)}$$

as we can check case by case: for $x^0 > y^0$:

$$\left[\phi^+(x), \phi^-(y)\right] = \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3q}{(2\pi)^3} \frac{1}{\sqrt{2E_{\vec{p}}}} \frac{1}{\sqrt{2E_{\vec{q}}}} e^{-ip\cdot x + iq\cdot y} \underbrace{\left[a_{\vec{p}}, a^+_{\vec{q}}\right]}_{(2\pi)^3 \delta^3(\vec{p}-\vec{q})}$$

$$= \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\vec{p}}} e^{-ip\cdot(x-y)} \Big|_{p^0 = E_{\vec{p}}} = D_F(x-y)$$

The same calculation w/ $y^0 > x^0$ gives us the opposite phase, as desired.

To summarize, we have

$$T\{\phi(x)\phi(y)\} = N\{\phi(x)\phi(y) + \overbrace{\phi(x)\phi(y)}\}$$

This generalizes (by induction) to products of more fields. The result is Wick's theorem:

$$T\{\phi(x_1)\phi(x_2)\ldots\phi(x_n)\}$$

$$= N\{\phi(x_1)\phi(x_2)\ldots\phi(x_n) + (\text{all possible contractions})\}$$

where all possible contractions means every way of pairing the fields. E.g., for $n=4$, writing $\phi(x_n) = \phi_n$:

$$T\{\phi_1\phi_2\phi_3\phi_4\} = N\{\phi_1\phi_2\phi_3\phi_4$$

$$+ \overbrace{\phi_1\phi_2}\phi_3\phi_4 + \overbrace{\phi_1\phi_3}\phi_2\phi_4 + \overbrace{\phi_1\phi_4}\phi_2\phi_3$$

$$+ \phi_1\overbrace{\phi_2\phi_3}\phi_4 + \phi_1\overbrace{\phi_2\phi_4}\phi_3 + \phi_1\phi_2\overbrace{\phi_3\phi_4}$$

$$+ \overbrace{\phi_1\phi_2}\overbrace{\phi_3\phi_4} + \overbrace{\phi_1\phi_3}\overbrace{\phi_2\phi_4} + \overbrace{\phi_1\phi_4}\overbrace{\phi_2\phi_3}$$

And the convention is that two non-adjacent operators contracted still results in a Feynman propagator:

$$N(\overbrace{\phi_1 \phi_2 \phi_3} \phi_4) = D_F(x_1 - x_3) N(\phi_2 \phi_4)$$

Now, taking the vacuum expectation value, any normal ordered product vanishes: $\langle 0 | N(\phi_1 \phi_2 \ldots) | 0 \rangle = 0$!

→ Only terms which are fully contracted give a nonzero result!

⇒ for four fields:

$$\langle 0 | T\{\phi_1 \phi_2 \phi_3 \phi_4\} | 0 \rangle$$

$$= \langle 0 | \overbrace{\phi_1 \phi_2} \overbrace{\phi_3 \phi_4} + \overbrace{\phi_1 \phi_2 \phi_3 \phi_4} + \overbrace{\phi_1 \phi_2 \phi_3 \phi_4} | 0 \rangle$$

$$= D_F(x_1 - x_2) D_F(x_3 - x_4) + D_F(x_1 - x_3) D_F(x_2 - x_4)$$

$$+ D_F(x_1 - x_4) D_F(x_2 - x_3)$$

We now know how to simplify terms in the expansion. It's time to make our lives a lot simpler by introducing Feynman diagrams.